

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse);library(lubridate);library(rvest);library(here);library(ggplot2)
here()
```

```
## [1] "/home/guest/R/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
url =
  'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024'
webpage = read_html(url)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- PWSID
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
WSN = webpage %>%
  html_node('div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()

PWSID = webpage %>%
  html_node('td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()

Ownership = webpage %>%
  html_node('div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()

MDU = webpage %>%
  html_nodes('th~ td+ td') %>%
  html_text()
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc. . .

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4
#indexing the month of MDU_mgd data
row = 4; col = 3
index = seq(1, row * col, by = row) + rep(0:(row - 1), each = col)
```

```

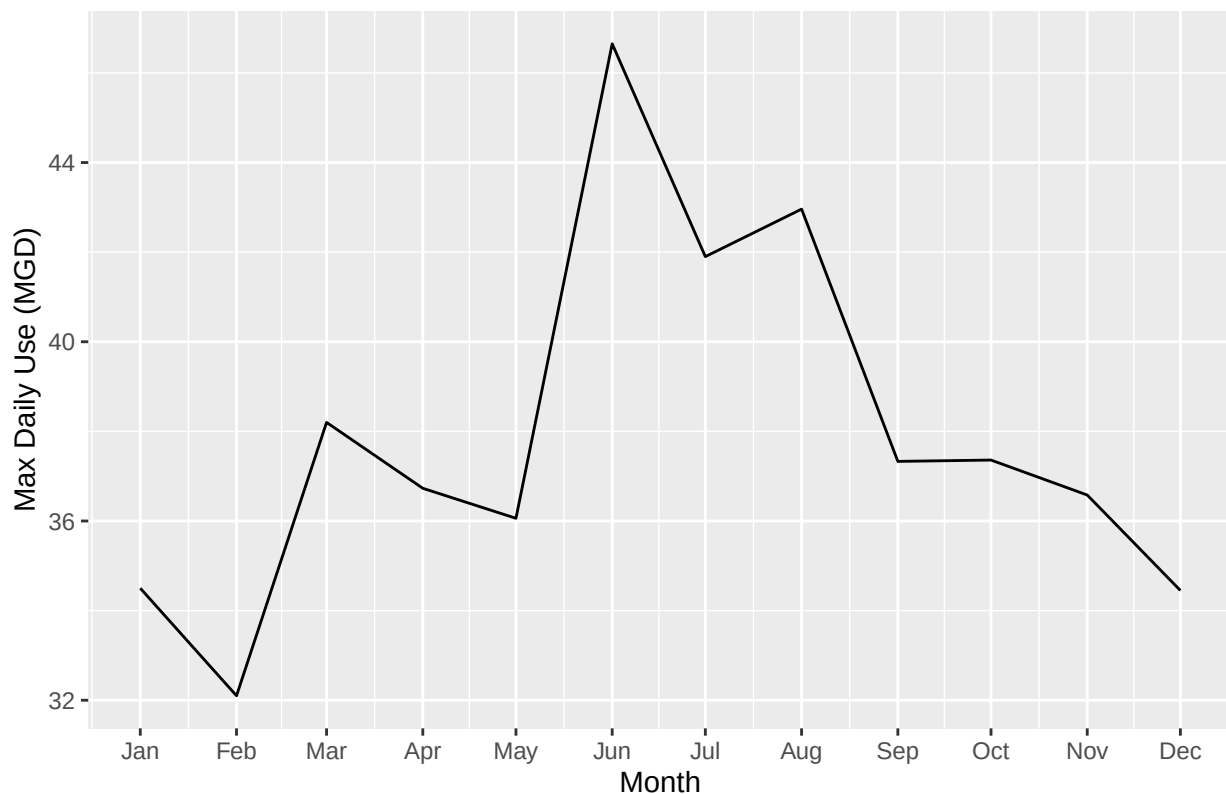
df = data.frame("Month" = month(index),
               "Year" = rep(2024,12),
               "WaterSystemName" = rep(WSN,12),
               "PWSID" = rep(PWSID, 12),
               "Ownership" = rep(Ownership, 12),
               "Max_Day_Use_mgd" = as.numeric(MDU)
               )
df$Date = as.Date(paste(df$Month, "01", df$Year, sep = "-"), format = "%m-%d-%Y")

#5

ggplot(df, aes(x = Date, y = Max_Day_Use_mgd)) +
  geom_line() +
  scale_x_date(
    date_labels = "%b",
    date_breaks = "1 month",
    limits = as.Date(c("2024-01-01", "2024-12-01"))
  ) +
  labs(
    title = "Maximum Daily Withdrawals (MGD) in Durham, 2024",
    x = "Month",
    y = "Max Daily Use (MGD)"
  )

```

Maximum Daily Withdrawals (MGD) in Durham, 2024



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a

function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
MDU_df = function(PWSID, year){
  base_url = 'https://www.ncwater.org/WUDC/app/LWSP/report.php?'
  scrape_url = paste0(base_url, 'pwsid=', PWSID, '&year=', year)
  webpage = read_html(scrape_url)

  WSN = webpage %>%
    html_node('div+ table tr:nth-child(1) td:nth-child(2)') %>%
    html_text()
#scraping
  PWSID = webpage %>%
    html_node('td tr:nth-child(1) td:nth-child(5)') %>%
    html_text()
  Ownership = webpage %>%
    html_node('div+ table tr:nth-child(2) td:nth-child(4)') %>%
    html_text()
  MDU = webpage %>%
    html_nodes('th~ td+ td') %>%
    html_text()
#creating df
  row = 4; col = 3
  index = seq(1, row * col, by = row) + rep(0:(row - 1), each = col)

  df = data.frame("Month" = month(index),
                  "Year" = rep(year, 12),
                  "WaterSystemName" = rep(WSN, 12),
                  "PWSID" = rep(PWSID, 12),
                  "Ownership" = rep(Ownership, 12),
                  "Max_Day_Use_mgd" = as.numeric(MDU)
                )
  df$Date = as.Date(paste(df$Month, "01", df$Year, sep = "-"),
                    format = "%m-%d-%Y")
  return(df)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
Durham2020 = MDU_df('03-32-010', 2020)
str(Durham2020)
```

```
## 'data.frame':    12 obs. of  7 variables:
## $ Month         : num  1 5 9 2 6 10 3 7 11 4 ...
```

```
## $ Year          : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ WaterSystemName: chr  "Durham" "Durham" "Durham" "Durham" ...
## $ PWSID         : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Ownership     : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Max_Day_Use_mgd: num  36 37 41.7 32 40.6 ...
## $ Date          : Date, format: "2020-01-01" "2020-05-01" ...
```

8. Use the function above to extract data for Cary (PWSID = ~~'03-32-010'~~'03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

#8

```
Cary2020 = MDU_df('03-92-020', 2020)
str(Cary2020)
```

```
## 'data.frame': 12 obs. of 7 variables:
## $ Month      : num  1 5 9 2 6 10 3 7 11 4 ...
## $ Year       : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ WaterSystemName: chr  "Cary" "Cary" "Cary" "Cary" ...
## $ PWSID      : chr  "03-92-020" "03-92-020" "03-92-020" "03-92-020" ...
## $ Ownership   : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Max_Day_Use_mgd: num  21.9 23.2 25.5 17.7 26.2 ...
## $ Date       : Date, format: "2020-01-01" "2020-05-01" ...
```

9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

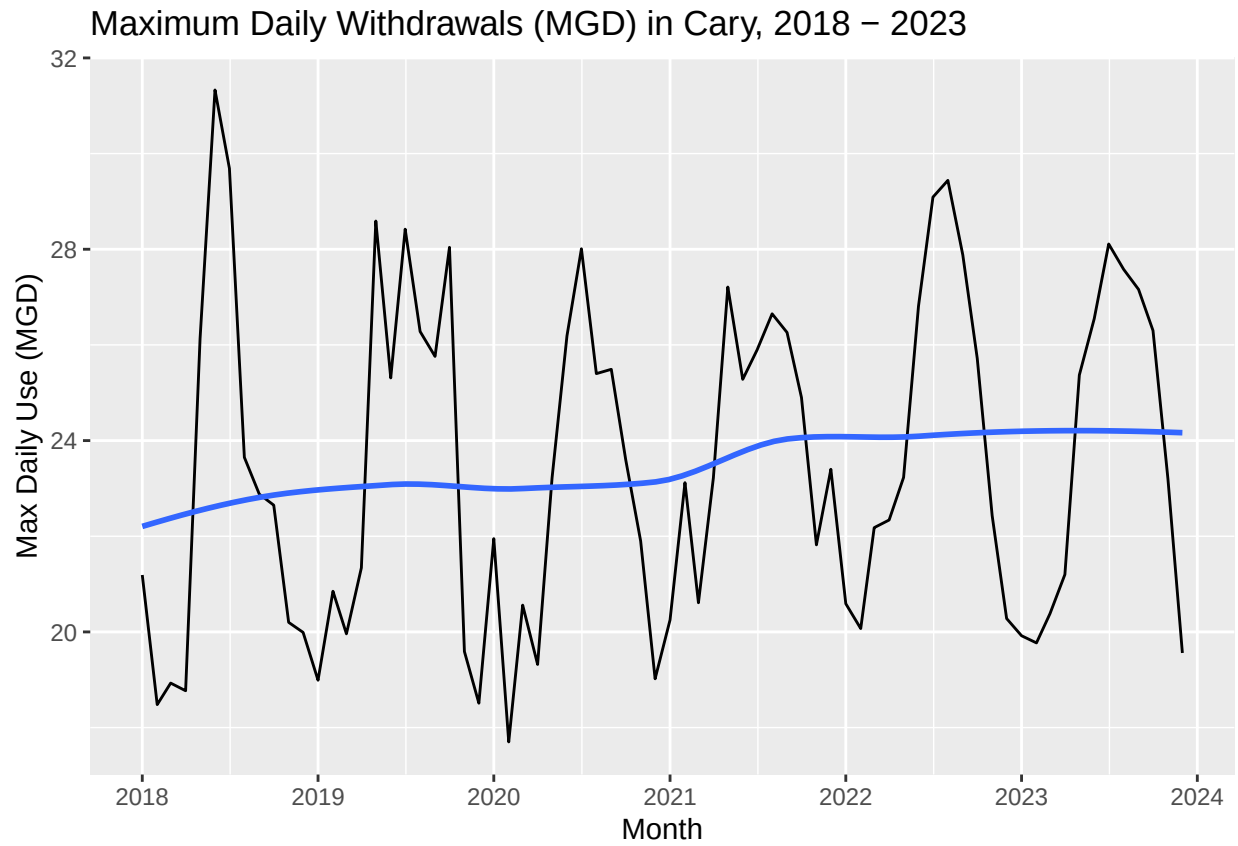
TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

#9

```
years = rep(2018:2023)
PWSID = '03-92-020'
Cary_2018_2023 = lapply(X = years,
                        FUN = MDU_df,
                        PWSID = '03-92-020'
                        ) %>%
  bind_rows()

ggplot(Cary_2018_2023, aes(x = Date, y = Max_Day_Use_mgd)) +
  geom_line() +
  geom_smooth(method='loess', se=FALSE) +
  scale_x_date(
    date_labels = "%Y",
    date_breaks = "1 year",
    limits = as.Date(c("2018-01-01", "2023-12-01"))
  ) +
  labs(
    title = "Maximum Daily Withdrawals (MGD) in Cary, 2018 - 2023",
    x = "Month",
    y = "Max Daily Use (MGD)"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: There's a slight upward tendency in overall water usage from 2018 through 2023, suggesting a small increase over time amidst significant seasonal fluctuations.